

Name

ID#

Math 150 Spring 2025 Final Exam

- 1) Evaluate the limit two ways
- a) Describe exactly what the limit represents in calculus terms.
- b) Use L'Hôpital's rule if possible (explain why or why you are not able to use the rule)

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos(x)} - 1}{x - \frac{\pi}{2}} \quad \frac{0}{0} \quad \frac{\frac{d}{dx} e^{\cos(x)} - 1}{\frac{d}{dx} x - \frac{\pi}{2}} = \frac{e^{\cos(x)} \cdot (-\sin(x))}{1}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{-e^{\cos(x)} \sin(x)}{1} = -e^{\cos(\frac{\pi}{2})} \sin(\frac{\pi}{2}) = -e^0 \cdot 1 = -1$$

$$\textcircled{1} \quad 3/5$$

$$\textcircled{2} \quad 5/5$$

$$\textcircled{3} \quad 5/5$$

last of 3

$$\begin{array}{l} \textcircled{5} \quad 6/5 \\ \textcircled{6} \quad 2/5 \\ \textcircled{7} \quad 5/5 \\ \textcircled{8} \quad 2/5 \end{array}$$

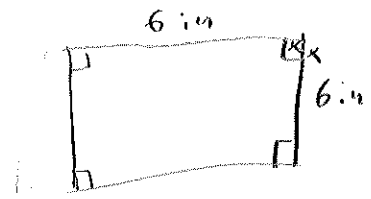
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$$\textcircled{9} \quad 5/5$$

$$\textcircled{10} \quad 5/5$$

Remaining sums geometry ✓

$$\frac{3 + 5 + 5 + 5 + 2 + 5 + 5 + 6}{40} = \frac{35}{40}$$



2) A sheet of paper measuring 6in by 6in is constructed into an open top box by removing four corners. Find the exact dimensions of the box with maximum volume. State the objective function and interval of interest.

$$\begin{aligned}\text{height} &= x \\ \text{width} &= 6 - 2x \\ \text{length} &= 6 - 2x\end{aligned}$$

Volume

$$\begin{aligned}V(x) &= x(6 - 2x)(6 - 2x) = x(6 - 2x)^2 \\ &= x(36 - 24x + 4x^2)\end{aligned}$$

$$V'(x) = 12x^2 - 48x + 36 = 4x^3 - 24x^2 + 36x$$

Interval of interest: $0 < x < 3$
 $x \in (0, 3)$

$$V'(x) = 12x^2 - 48x + 36 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3)$$

$$x = 1 \quad x = 3$$

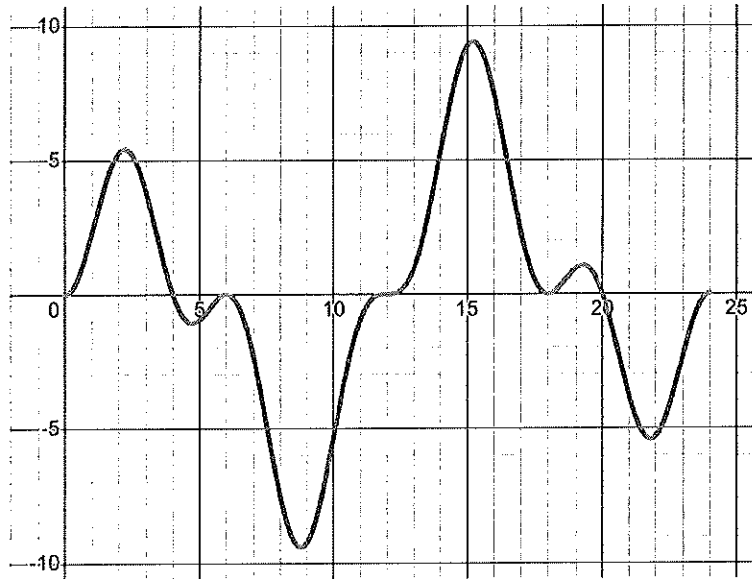
$$\begin{aligned}V(1) &= 4(1)^3 - 24(1)^2 + 36(1) \\ &= 4 - 24 + 36 = 16\end{aligned}$$

$$\begin{aligned}\text{length} &= 6 - 2(1) = 4 \text{ in} \\ \text{width} &= 6 - 2(1) = 4 \text{ in} \\ \text{height} &: 1 \text{ in}\end{aligned}$$

$$P(t) = E'(t)$$

4) Given the graph of $E(t)$ over a 24 hour period below, and that power $P(t) = E'(t)$.

Graph power $P(t)$ and estimate the time(s) is the power is maximized and minimized.



0-24 in 24 hrs

$$2\sin(2x) = 2\sin x \cos x \quad (\cos x)(2\sin x - 1) = 0$$

$$2\sin x \cos x = \cos x$$

$$\sin x \cos x - \cos x = 0$$

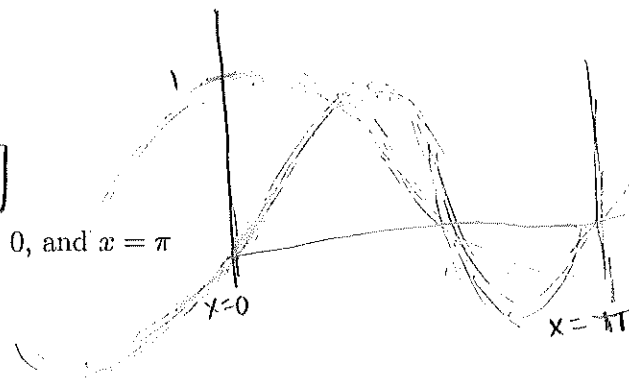
$$[0, \pi]$$

6) Find the area between the curves $y = \sin(2x)$, $y = \cos(x)$, $x = 0$, and $x = \pi$

$$\cos x = 0 \quad x = \frac{\pi}{2} \quad 2\sin x - 1 = 0$$

$$\sin x = \frac{1}{2} \quad x = \frac{\pi}{6}$$

$$x = \frac{5\pi}{6}$$



$$A = \int_0^{\pi/6} (\cos(x) - \sin(2x)) dx + \int_{\pi/6}^{\pi/2} (\sin(2x) - \cos(x)) dx$$

$$+ \int_{\pi/2}^{5\pi/6} (\cos(x) - \sin(2x)) dx$$

$$+ \int_{5\pi/6}^{\pi} (\sin(2x) - \cos(x)) dx$$

$$\int (\cos(x) - \sin(2x)) dx = \sin(x) + \frac{1}{2} \cos(2x) + C$$

$$\int (\sin(2x) - \cos(x)) dx = -\frac{1}{2} \cos(2x) - \sin(x) + C$$

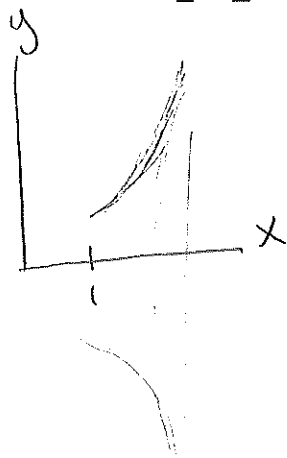
$$\int_0^{\pi/6} (\cos(x) - \sin(2x)) dx = \sin(x) + \frac{1}{2} \cos(2x) \Big|_0^{\pi/6}$$

$$\left(\sin\left(\frac{\pi}{6}\right) + \frac{1}{2} \cos\left(\frac{\pi}{3}\right) \right) - \left(\sin(0) + \frac{1}{2} \cos(0) \right)$$

$$= \left(\frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \right) - \left(0 + \frac{1}{2} \cdot 1 \right) = \underline{\underline{\frac{1}{4}}}$$

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7) Calculate the surface area of the surface obtained by revolving the curve $y = \frac{x^3}{3}$ around the x -axis from $1 \leq x \leq 2$.



$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{x^3}{3} \right) = x^2 \quad A = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$y = \frac{x^3}{3} \quad \frac{dy}{dx} = x^2 \quad a=1 \quad b=2$$

$$A = 2\pi \int_1^2 \frac{x^3}{3} \sqrt{1 + (x^2)^2} dx$$

$$= 2\pi \int_1^2 \frac{x^3}{3} \sqrt{1 + x^4} dx \quad u = 1 + x^4$$

$$\frac{du}{dx} = 4x^3$$

$$dx = \frac{du}{4x^3}$$

$$x=1, u=1+1^4=2$$

$$x=2, u=1+2^4=1+16=17$$

$$A = 2\pi \int_2^{17} \frac{x^3}{3} \sqrt{u} \frac{du}{4x^3} = \frac{2\pi}{12} \int_2^{17} \sqrt{u} du = \frac{\pi}{6} \int_2^{17} u^{1/2} du$$

$$A = \frac{\pi}{6} \left(\frac{2}{3} u^{3/2} \right) \Big|_2^{17} = \frac{\pi}{6} \cdot \frac{2}{3} (u^{3/2}) \Big|_2^{17}$$

$$\frac{\pi}{9} (17^{3/2} - 2^{3/2})$$

$$A = \frac{\pi}{9} (17\sqrt{17} - 2\sqrt{2}) =$$

$$-4.5 \rightarrow 4.5$$

$$A = \pi r^2$$

$$\frac{1}{2} \pi r^2$$

8) Find the exact value of the integral $\int_{-5}^5 \sqrt{25-x^2} dx$ using integration techniques and geometry.

Use Riemann Sums to approximate the integral with 10 rectangles and use Sigma notation (draw the rectangles on the graph).

How many rectangles are necessary to approximate to within 0.001 of the exact value?

$$y = \sqrt{25-x^2} \quad r=5$$

$$x = -5 \quad \text{to} \quad x = 5$$

$$a \quad b$$

$$n=10$$

$$\Delta x = \frac{b-a}{n}$$

$$\frac{5-(-5)}{10} = 1$$

$$\frac{1}{2} \pi (5^2) = \frac{25\pi}{2}$$

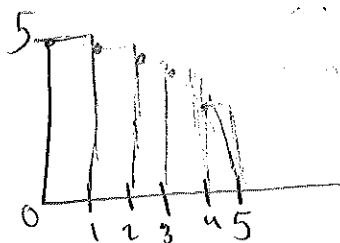
$$x_1 = \sqrt{25 - (0)^2} = 5$$

$$x_2 = \sqrt{25 - (0.5)^2} = 4.9609$$



$$\sum_{k=1}^{10} f(x_k) \Delta x = \frac{1}{10} (x_1 + x_2 + x_3 + x_4 + x_5 + \dots + x_{10})$$

$$= 32.96311036$$

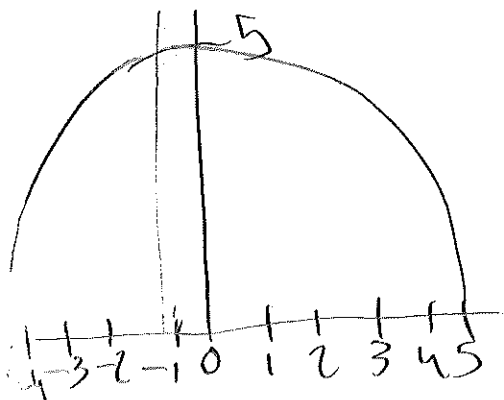


$$f(x_k) = \sqrt{25 - (x_k)^2}$$

$$k=0 \quad x=1$$

$$\sum_{k=1}^{10} \sqrt{25 - (-5 + (k-0.5))^2}$$

$$\sqrt{25 - (4.5)^2} \quad k=1$$



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9) Let $y = \tan^{-1}(x)$. Implicitly differentiate to derive the derivative $\frac{d}{dx}(\tan^{-1}(x))$.

$$y = \tan^{-1}(x) \quad \frac{d}{dx}(\tan(y)) = \frac{d}{dx}(x)$$

$$\tan(y) = x$$

$$\sec^2(y) \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

arctan

$$\frac{dy}{dx} = \frac{1}{\sec^2(y)}$$

$$\sec^2(y) = 1 + \tan^2(y)$$

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2(y)}$$

9b) Let $y = \cosh^{-1}(x)$. Implicitly differentiate to derive the derivative $\frac{d}{dx}(\cosh^{-1}(x))$.

$$x = \cosh(y) \quad \frac{d}{dx}(x) = \frac{d}{dx}(\cosh(y))$$

$$\sinh(y) \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sinh(y)}$$

$$\cosh^2(y) - \sinh^2(y) = 1$$

$$\sinh^2(y) = \cosh^2(y) - 1$$

$$\sinh(y) = \sqrt{\cosh^2(y) - 1}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2(y) - 1}}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$$

$$x = y^2 \quad y = x^2$$

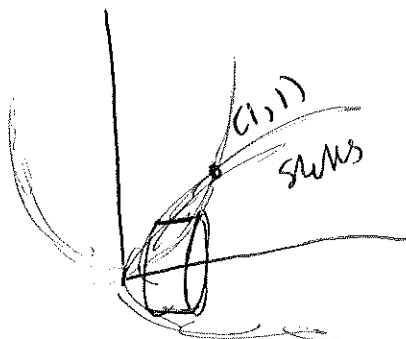
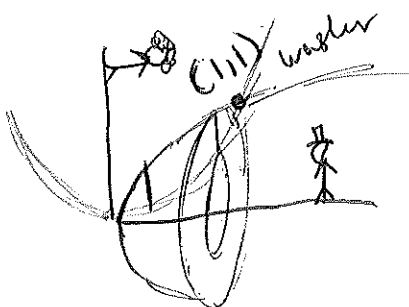
$$y = x^2 \quad x = y^2$$

$$x = (x^2)^2 = x^4 \quad x^4 - x = 0$$

$$x(x^3 - 1) = 0 \quad \underline{x=0} \quad \underline{x=1}$$

$$(0,0) \quad (1,1)$$

10) Find the volume generated by revolving the region bounded by the curves $y = x^2$ and $x = y^2$ about the x -axis first using washers, then use shells. Show the two integrals are equal.



$$V = \pi \int_0^1 ((\sqrt{x})^2 - (x^2)^2) dx$$

$$= \pi \int_0^1 (x - x^4) dx$$

$$V = \pi \left(\frac{x^2}{2} - \frac{x^5}{5} \right) \Big|_0^1$$

$$= \pi \left(\frac{1}{2} - \frac{1}{5} \right) = \pi \left(\frac{5-2}{10} \right) = \frac{3\pi}{10}$$

$$V = 2\pi \int_0^1 y (\sqrt{y} - y^2) dy = 2\pi \int_0^1 (y^{3/2} - y^3) dy$$

$$= 2\pi \left(\frac{y^{5/2}}{5/2} - \frac{y^4}{4} \right) \Big|_0^1 = 2\pi \left(\frac{2y^{5/2}}{5} - \frac{y^4}{4} \right) \Big|_0^1$$

$$= 2\pi \left(\frac{2}{5} - \frac{1}{4} \right)$$

$$2\pi \left(\frac{8-5}{20} \right) = 2\pi \left(\frac{3}{20} \right)$$

$$= \frac{3\pi}{10}$$